REVTEX4

First Author and Second Author

My Institution

Third Author

Other Institution

(Date textdate; Received textdate; Revised textdate; Accepted textdate; Published textdate)

Abstract

Shell document for REVTEX 4.

Contents

Part I

Dual Space Theory

I. INTRODUCTION

This REVTeX document preserves the complete verbatim discussion about vector space duals, Schauder bases, biorthogonal systems, and their applications to distribution theory. Each section maintains the original question-answer format while adhering to REVTeX conventions.

II. INITIAL QUESTION: TRIPLE DUAL ISOMORPHISM

Question

Is the triple dual of a vector space isomorphic to the dual in infinite dimension?

Answer

Excellent question. The short answer is: **No, in infinite dimensions, the triple dual V^{***} is not necessarily isomorphic to the dual V^*.**

In fact, the relationship between a vector space and its duals becomes much more complex and nuanced in infinite dimensions. Let's break down why.

A. Finite-Dimensional Case

In finite dimensions, everything works very neatly. Let V be a finite-dimensional vector space over a field $\mathbb{F}$.

1. **Dual Space (V^*)**: The space of all linear functionals $f : V \to \mathbb{F}$. It's a standard theorem that $\dim(V^*) = \dim(V)$.

2. **Double Dual (V^{**})**: The space of all linear functionals $\phi : V^* \to \mathbb{F}$. There is a **natural map** $\iota : V \to V^{**}$ defined by $\iota(v)(f) = f(v)$ for all $f \in V^*$. This map is always injective. Since the dimensions are equal, ι is an **isomorphism**. We say V is **reflexive** ($V \cong V^{**}$).

3. **Triple Dual (V^{***})**: We can use the natural isomorphism from V to V^{**} to create an isomorphism between V^* and V^{***}.

   - Start with the natural map $\iota : V \to V^{**}$.
   - Take its dual map: $\iota^* : (V^{**})^* \to V^*$. Since $(V^{**})^* = V^{***}$, we have $\iota^* : V^{***} \to V^*$.
   - Because ι is an isomorphism, its dual map ι^* is also an isomorphism.
   - Therefore, $V^{***} \cong V^*$.

So, in finite dimensions, $V \cong V^{**}$ and $V^* \cong V^{***}$.

B. Infinite-Dimensional Case

In infinite dimensions, the natural map $\iota : V \to V^{**}$ is still injective, but it is **not surjective**. This means V is a *proper* subspace of V^{**}. We say V is **not reflexive**. This failure of reflexivity breaks the nice chain of isomorphisms.

- V^{**} is "larger" than V.
- $V^{***} = (V^{**})^*$ is the dual of this "larger" space.

III. HIGHER ORDER DUALS

Question

Do the higher order duals ever become isomorphic?

Answer

Another excellent and deep question. The answer is a fascinating mix of "no," "sometimes," and "it depends on your set theory."

Let's denote the process of taking duals repeatedly:

- $V^{(0)} = V$
- $V^{(1)} = V^*$
- $V^{(2)} = V^{**}$
- $V^{(3)} = V^{***}$
- ... and so on.

We want to know if for some n, $V^{(n)} \cong V^{(n+k)}$ for all $k \geq 0$, creating a "cycle" of isomorphisms.

Here's the breakdown of what can happen.

A. General Case: No Stabilization

For a "general" infinite-dimensional vector space V (using the standard axioms of Zermelo-Fraenkel set theory with the Axiom of Choice, ZFC), the sequence of duals **grows forever in size**.

- Each dual space is strictly larger than the previous one in terms of algebraic dimension (a measure of the "number of basis vectors").
- More precisely, if $\dim(V)$ is an infinite cardinal number κ, then $\dim(V^*) = |\mathbb{F}|^\kappa$, which is strictly larger than κ. Applying the dual again gives an even larger dimension $|\mathbb{F}|^{|\mathbb{F}|^\kappa}$, and so on.
- Since isomorphic spaces must have the same algebraic dimension, this means: $V \ncong V^* \ncong V^{**} \ncong V^{***} \ncong \ldots$

In this general case, the sequence of duals **never becomes isomorphic**. Each step gives you a fundamentally larger and more complex space.

B. Special Case: Reflexive Spaces

A Banach space X is called **reflexive** if the natural map $\iota : X \to X^{**}$ is an isomorphism.

IV. NATURAL MAP VS ISOMORPHISM

Question

I thought isomorphism meant injective and surjective so how can the embedding not be surjective?

Answer

You have hit on a crucial and subtle point that is the source of a great deal of depth and confusion in this area. Your understanding is correct, and the resolution lies in the distinction between two different concepts:

1. **Isomorphism**
2. **The Natural Embedding (or Canonical Map)**

Let's clarify this.

A. Isomorphism

An isomorphism between two vector spaces V and W is a linear map $T : V \to W$ that is both **injective** and **surjective**.

- If such a map exists, we say V and W are **isomorphic** and write $V \cong W$.
- The key point is that the existence of *some* bijective linear map is enough to declare two spaces isomorphic. It doesn't have to be a "natural" or "obvious" one.

B. Natural Embedding

For any vector space V, there is a specific, "canonical" way to map it into its double dual. This is the map $\iota : V \to V^{**}$ defined by:

$$[\iota(v)](f) = f(v) \quad \text{for all } v \in V, f \in V^*.$$

Let's break down its properties:

- **It is always injective and linear.** You can prove that if $\iota(v) = 0$, then $f(v) = 0$ for every linear functional f, which forces $v = 0$.
- **It is not always surjective.** Whether it's surjective or not depends entirely on the dimension of V.

C. Two Different Maps

The magic and the confusion arise because we are often talking about **two different maps** connecting V and V^{**}.

V. BIORTHOGONAL INJECTIONS

Question

Can one define injections from a vector space to its dual by biorthogonality?

Answer

Excellent question! The short answer is **yes, but with very strong caveats**. The concept of biorthogonality is a powerful tool for defining injections from a vector space to its dual, but it works well only in specific, well-structured situations.

Let's explore this in detail.

A. Finite-Dimensional Case: Perfect Biorthogonality

In finite dimensions, biorthogonality gives us a perfect way to create isomorphisms (not just injections) between V and V^*.

Let V be finite-dimensional with basis $\{e_1, \ldots, e_n\}$. We can define the **dual basis** $\{f_1, \ldots, f_n\} \subset V^*$ by:

$$f_i(e_j) = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

The biorthogonality map $B : V \to V^*$ is defined by linearly extending:

$$B(e_i) = f_i$$

This map B is an **isomorphism** (bijective). Every vector $v = \sum v_i e_i$ maps to the functional $B(v) = \sum v_i f_i$.

Key point: In finite dimensions, biorthogonal systems always exist and give us isomorphisms.

B. Infinite-Dimensional Challenge

In infinite dimensions, things become much more difficult. The main issues are:

1. Hamel Bases and Their Limitations

Every vector space has a Hamel basis (by the Axiom of Choice), but:

- Hamel bases are uncountable for infinite-dimensional spaces
- The dual elements defined via biorthogonality won't necessarily be continuous/bounded

- The resulting map to the algebraic dual might not land in the topological dual we care about

2. Minimal Requirement: A Biorthogonal System

VI. HAMEL VS SCHAUDER BASES

Question

In an infinite separable Hilbert space the basis is countable is this is not a Hamel basis?

Answer

Exactly right! This is a crucial and often confusing point. In an infinite-dimensional separable Hilbert space:

- The **orthonormal basis** (like the standard basis in ℓ^2) is **countable**
- But a **Hamel basis** (a linearly independent set whose finite linear combinations span the space) is **uncountable**

Let me explain why this happens and why it matters.

A. Definitions Clarified

1. Hamel Basis (Algebraic Basis)

A Hamel basis is a set $\{e_i\}_{i\in I} \subset H$ such that:

1. Every finite subset is linearly independent
2. Every vector $x \in H$ can be written as a **finite linear combination** $x = \sum_{k=1}^{n} \alpha_k e_{i_k}$

2. Orthonormal Basis (Hilbert Space Basis)

A countable set $\{e_n\}_{n=1}^{\infty} \subset H$ such that:

1. $\langle e_i, e_j \rangle = \delta_{ij}$
2. Every vector $x \in H$ can be written as $x = \sum_{n=1}^{\infty} \alpha_n e_n$ (convergent infinite series)

B. Why a Hamel Basis Must Be Uncountable

Theorem: In any infinite-dimensional Banach space (including separable Hilbert spaces), every Hamel basis is uncountable.

Proof sketch: The Baire Category Theorem tells us that a complete metric space cannot be a countable union of nowhere dense sets. If a Hamel basis were countable, the space would be a countable union of finite-dimensional subspaces, which are nowhere dense in infinite dimensions. Contradiction.

Concrete example in ℓ^2:

- The standard orthonormal basis $\{(1, 0, 0, \dots), (0, 1, 0, \dots), \dots\}$ is countable

VII. SCHAUDER BASIS DEFINITION

Question

What is the definition of a Schauder basis?

Answer

A **Schauder basis** is a generalization of the concept of a basis from linear algebra to infinite-dimensional Banach spaces (and more generally, topological vector spaces). It allows us to represent every vector in the space as an **infinite linear combination** of basis elements.

A. Formal Definition

Let X be a Banach space. A sequence $\{e_n\}_{n=1}^{\infty}$ in X is called a **Schauder basis** for X if for every vector $x \in X$, there exists a **unique** sequence of scalars $\{\alpha_n\}_{n=1}^{\infty}$ such that:

$$x = \sum_{n=1}^{\infty} \alpha_n e_n$$

where the convergence is in the norm topology of X, i.e.:

$$\lim_{N\to\infty} \left\| x - \sum_{n=1}^{N} \alpha_n e_n \right\| = 0$$

B. Key Properties and Consequences

1. Uniqueness of Representation

The uniqueness requirement means that if $\sum_{n=1}^{\infty} \alpha_n e_n = \sum_{n=1}^{\infty} \beta_n e_n$, then $\alpha_n = \beta_n$ for all n.

2. Coordinate Functionals

For a Schauder basis $\{e_n\}$, there exist unique continuous linear functionals $\{f_n\} \subset X^*$ called the **coordinate functionals** or **biorthogonal functionals** such that:

$$f_n(e_m) = \delta_{nm} = \begin{cases} 1 & \text{if } n = m \\ 0 & \text{if } n \neq m \end{cases}$$

and for every $x \in X$, we have:

$$x = \sum_{n=1}^{\infty} f_n(x) e_n$$

3. Basis Constant

VIII. DIFFERENT BASES GIVE DIFFERENT INJECTIONS

Question

Are the injections defined by biorthogonality different for different Schauder bases?

Answer

Yes, absolutely! The injections defined by biorthogonality can be (and usually are) **completely different** for different Schauder bases of the same Banach space.

Let me explain why this happens and give some concrete examples.

A. General Construction

For a Schauder basis $\{e_n\}$ with biorthogonal functionals $\{f_n\} \subset X^*$, we can define an injection $\Phi : X \to X^*$ by:

$$\Phi\left(\sum_{n=1}^{\infty} a_n e_n\right) = \sum_{n=1}^{\infty} a_n f_n$$

This map sends each basis vector e_n to its corresponding dual functional f_n.

B. Why Different Bases Give Different Injections

1. Different Coordinate Systems

Different Schauder bases represent vectors using different "coordinate systems":

- If $x = \sum a_n e_n$ in basis $\{e_n\}$
- And $x = \sum b_n g_n$ in basis $\{g_n\}$
- Then typically $a_n \neq b_n$

Since $\Phi_{\{e_n\}}(x) = \sum a_n f_n$ and $\Phi_{\{g_n\}}(x) = \sum b_n h_n$, these are different functionals acting on X.

2. Different Ranges

The image of each injection is the closed linear span of the corresponding biorthogonal functionals:

- $\text{range}(\Phi_{\{e_n\}}) = \overline{\text{span}}\{f_n\}$
- $\text{range}(\Phi_{\{g_n\}}) = \overline{\text{span}}\{h_n\}$

These can be different subspaces of X^*.

C. Concrete Example in ℓ^p

IX. SPACES WITH SCHAUDER BASES

Question

Is there a name for Banach spaces that have Schauder bases?

Answer

Yes, there are several important names and concepts related to Banach spaces with Schauder bases:

A. Spaces with the "Basis Property"

A Banach space is said to have the **basis property** if it admits a Schauder basis. However, this terminology is somewhat informal.

B. More Precisely: Spaces with the "Approximation Property"

There's a fundamental hierarchy:

- **Schauder basis** $\Rightarrow$ **Approximation property**
- But **Approximation property** $\not\Rightarrow$ **Schauder basis**

The **approximation property** means that compact operators can be approximated by finite-rank operators. Every space with a Schauder basis has the approximation property, but the converse is false (as shown by Enflo's famous counterexample).

C. Important Subclasses

1. Spaces with Unconditional Bases

These are Banach spaces that admit an **unconditional Schauder basis** - where every convergent rearrangement of the basis representation converges to the same vector. Examples include:

- ℓ^p spaces $(1 \leq p < \infty)$
- c_0
- $L^p[0,1]$ for $1 < p < \infty$

2. Spaces with Boundedly Complete Bases

A basis $\{e_n\}$ is **boundedly complete** if whenever $\sup_N \|\sum_{n=1}^N a_n e_n\| < \infty$, then $\sum_{n=1}^\infty a_n e_n$ converges.

3. Spaces with Shrinking Bases

X. DISTRIBUTION THEORY AND TVS

Question

Is distribution theory based on TVS and their duals?

Answer

Yes, absolutely! Distribution theory is fundamentally built on the theory of topological vector spaces (TVS) and their duals. This is not just a minor connection – it's the very foundation upon which distribution theory is constructed.

Let me explain how this works.

A. Basic Setup

1. Test Functions

We start with spaces of **test functions**:

- $\mathcal{D}(\Omega) = C_c^\infty(\Omega)$ (compactly supported smooth functions)
- $\mathcal{S}(\mathbb{R}^n)$ (Schwartz space of rapidly decreasing functions)

These are **topological vector spaces** with carefully chosen topologies that make them:

- Complete
- Hausdorff
- Locally convex
- But **not** normable

2. Distributions as Duals

Distributions are defined as **continuous linear functionals** on these spaces:

- $\mathcal{D}'(\Omega) = \mathcal{D}(\Omega)^*$ (distributions)
- $\mathcal{S}'(\mathbb{R}^n) = \mathcal{S}(\mathbb{R}^n)^*$ (tempered distributions)

So distributions literally **are** the dual spaces of certain TVS!

B. TVS Structure Matters Crucially

1. Why the Topology is Carefully Chosen

The topology on test function spaces is designed so that:

- Differentiation is continuous

XI. BASES IN DISTRIBUTION SPACES

Question

Do these spaces have Schauder bases?

Answer

Excellent question! The answer reveals a fascinating and nuanced picture in functional analysis. The short answer is:

- **Yes** for some test function spaces (with careful interpretation)
- **No** for others
- **It's complicated** and depends on exactly which space and what kind of basis we mean

Let me break this down carefully.

A. The Schwartz Space $\mathcal{S}(\mathbb{R}^n)$

The Schwartz space of rapidly decreasing smooth functions **does have a kind of Schauder basis**, but it's not straightforward.

1. Hermite Basis

For $\mathcal{S}(\mathbb{R})$, the **Hermite functions** form a basis:

$$h_n(x) = (-1)^n e^{x^2/2} \frac{d^n}{dx^n}(e^{-x^2})$$

These are:

- Complete in $\mathcal{S}(\mathbb{R})$
- Orthonormal in $L^2(\mathbb{R})$
- Each $h_n \in \mathcal{S}(\mathbb{R})$

However, the convergence $\phi = \sum a_n h_n$ in the Schwartz space topology is much stronger than L^2 convergence.

2. Multi-dimensional Case

For $\mathcal{S}(\mathbb{R}^n)$, tensor products of Hermite functions work:

$$h_\alpha(x) = h_{\alpha_1}(x_1) \cdots h_{\alpha_n}(x_n)$$

This gives a Schauder basis for $\mathcal{S}(\mathbb{R}^n)$.